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Introduction

DTS is the native binary format used by the Torque engine to store shape, mesh and (optionally) sequence data. This document serves as a basic specification for the DTS file format. Very little context is given to the shape, mesh and sequence data fields read from the DTS file so it is recommended that this resource be read along with the Torque 3D source code.

The DTS file format stores most data in separate buffers for 8, 16 or 32-bit sized values in order to speed up loading on non-Intel platforms. A DTS file can be read as shown below; after the 3 data buffers have been created in memory, they are read in interleaved order as needed to extract the desired values (see Data Buffers).

Note: *that all values are stored as little-endian, so on non-Intel platforms you will have to handle perform endian conversion for 16 and 32-bit values.*

DTS Format

Name	# bytes	Data Type	Description
DTS version number	2	S16	DTS version of this file. Note that the current DTS version (for files saved by Torque 3D) is 26. Most DTS exporters and the ShowToolPro tool only read/write version 24.
Exporter version number	2	S16	For tracking version of the exporter that generated this file. Can usually be ignored.
sizeAll	4	S32	Size (in 32-bit words) of the 3 data buffers (sum of 8, 16 and 32-bit data buffers). The 32-bit data buffer starts at offset 0x0.

start16	4	S32	32-bit word offset into the memory buffer of the start of the 16-bit data.
start8	4	S32	32-bit word offset into the memory buffer of the start of the 8-bit data.
data32	size32	U32*	Data buffer for 32-bit values. Size (in bytes) is equal to start16*4
data16	size16	U16*	Data buffer for 16-bit values. Size (in bytes) is equal to (start8 - start16)*4
data8	size8	U8*	Data buffer for 8-bit values. Size (in bytes) is equal to (sizeAll - start8)*4
numSequences	4	S32	Number of sequences in the shape
sequences	numSequences * ?	Sequence	Array of numSequences Sequences - size of each sequence depends on which nodes and objects are affected etc. See Sequences for more information
matStreamType	1	S8	Type of material stream (text or binary). Should be 0x1 (binary) for DTS files
numMaterials	4	S32	Number of materials in the shape
matNames	numMaterials * ?	S32 and char[]	Names of the materials in the shape. Each name is stored as a 4-byte length followed by the N characters in the string (terminating NULL is not included in the length or N characters).
matFlags	numMaterials * 4	U32	Flags for each material*
matReflectanceMaps	numMaterials * 4	S32	Index of the material to use as a reflectance map for each material* (or -1 for none)
matBumpMaps	numMaterials * 4	S32	Index of the material to use as a bump map for each material* (or -1 for none)
matDetailMaps	numMaterials * 4	S32	Index of the material to use as a detail map for each material* (or -1 for none)
dummy	numMaterials * 4	S32	Dummy value. Only present in DTS v25.
matDetailScales	numMaterials * 4	F32	Detail scale for each material*
matReflectance	numMaterials * 4	F32	Reflectance value for each material*

***Note:** that all material settings (beside the name) are superceded by script Material definitions in T3D.*

Sequences

Sequences are stored in the DTS file as follows:

Name	**# bytes**	**Data Type**	**Description**
nameIndex	4	S32	The name of this sequence as in index into the names array
flags	4	U32	Sequence flags
numKeyframes	4	S32	Number of keyframes in this sequence
duration	4	F32	Duration of the sequence (in seconds)
priority	4	S32	Sequence priority

firstGroundFrame	4	S32	First ground transform keyframe in this sequence (index into the groundTranslations and groundRotation arrays)
numGroundFrames	4	S32	Number of ground transform keyframes in this sequence
baseRotation	4	S32	First node rotation keyframe in this sequence (index into the nodeRotations array)
baseTranslation	4	S32	First node translation keyframe in this sequence (index into the nodeTranslations array)
baseScale	4	S32	First node scale keyframe in this sequence (index into the nodeXXXScales arrays)
baseObjectState	4	S32	First object state keyframe in this sequence (index into the objectStates array)
baseDecalState	4	S32	First decal state keyframe in this sequence (index into the decalStates array). Note that DTS decals are deprecated, and this value should be 0.
firstTrigger	4	S32	First trigger in this sequence (index into the triggers array)
numTriggers	4	S32	Number of triggers in this sequence
toolBegin	4	F32	Value representing the start of this sequence in the exporting tool's timeline (can usually by ignored)
rotationMatters	?	BitSet	BitSet indicating which node rotations are animated by this sequence.
translationMatters	?	BitSet	BitSet indicating which node translations are animated by this sequence.
scaleMatters	?	BitSet	BitSet indicating which node scales are animated by this sequence.
decalMatters	?	BitSet	BitSet indicating which decal states are animated by this sequence. Note that DTS decals are deprecated.
iflMatters	?	BitSet	BitSet indicating which IFL materials are animated by this sequence.
visMatters	?	BitSet	BitSet indicating which object's visibility is animated by this sequence.
frameMatters	?	BitSet	BitSet indicating which mesh's verts are animated by this sequence.
matFrameMatters	?	BitSet	BitSet indicating which mesh's UV coords are animated by this sequence.

BitSets

BitSets are stored in the DTS file as:

Name	# bytes	Data Type	Description
dummy	4	S32	Dummy value
numWords	4	S32	Number of 32-bit values in the BitSet
bits	4 * numWords	U32	Array of numWords 32-bit values representing the bits in the BitSet

Data Buffers

The 3space shape data is extracted from the 3 data buffers by maintaining a pointer into each buffer, then reading from the appropriate buffer in order as shown in the table below. The code sample below demonstrates how an array of structures might be read from the 3 buffers.

```
struct
{
   S8 valueA[4];
   S32 valueB;
   S32 valueC;
   S16 valueD;
} Object;

...

#define checkGuard() assert((*buff32++ == guard32++) && (*buff16++ == guard16++)
 && (*buff8++ == guard8++))

...

numObjects = *buff32++;
objects.resize(numObjects);
for (S32 i = 0; i < numObjects; i++)
{
   for (S32 j = 0; j < 4; j++)
      objects[i].valueA[j] = *buff8++;

   objects[i].valueB = *buff32++;
   objects[i].valueC = *buff32++;
   objects[i].valueD = *buff16++;
}

checkGuard();
```

The GUARD entries indicate a buffer checkpoint that verifies the buffers are being accessed correctly. Each time the GUARD entry is encountered, an incrementing value should be read from each of the 3 buffers. For example, the first GUARD should read 32-bit 0x0, 16-bit 0x0 and 8-bit 0x0, the next GUARD should read 32-bit 0x1, 16-bit 0x1 and 8-bit 0x1 and so on.

Name	**Buffer**	**Data Type**	**Description**
numNodes	32-bit	S32	Number of nodes in the shape
numObjects	32-bit	S32	Number of objects in the shape
numDecals	32-bit	S32	Number of decals in the shape
numSubShapes	32-bit	S32	Number of subshapes in the shape
numIFLs	32-bit	S32	Number of IFL materials in the shape
numNodeRotations	32-bit	S32	Number of node rotation keyframes
numNodeTranslations	32-bit	S32	Number of node translation keyframes
numNodeUniformScales	32-bit	S32	Number of node uniform scale keyframes
numNodeAlignedScales	32-bit	S32	Number of node aligned scale keyframes
numNodeArbScales	32-bit	S32	Number of node arbitrary scale keyframes
numGroundFrames	32-bit	S32	Number of ground transform keyframes

numObjectStates	32-bit	S32	Number of object state keyframes
numDecalStates	32-bit	S32	Number of decal state keyframes
numTriggers	32-bit	S32	Number of triggers (all sequences)
numDetails	32-bit	S32	Number of detail levels in the shape
numMeshes	32-bit	S32	Number of meshes (all detail levels) in the shape
numNames	32-bit	S32	Number of name strings in the shape
smallestVisibleSize	32-bit	F32	Size of the smallest visible detai level
smallestVisibleDL	32-bit	S32	Index of the smallest visible detail level
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
radius	32-bit	F32	Shape bounding sphere radius
tubeRadius	32-bit	F32	Shape bounding cylinder radius
center	32-bit	Point3F { F32 x, F32 y, F32 z }	Center of the shape bounds
bounds	32-bit	BoxF { Point3F min, Point3F max }	Shape bounding box
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
nodes	32-bit	Node { S32 nameIndex, S32 parentIndex, S32 firstObject, S32 firstChild, S32 nextSibling }	Array of numNodes Nodes
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
objects	32-bit	Object { S32 nameIndex, S32 numMeshes, S32 startMeshIndex, S32 nodeIndex, S32 nextSibling, S32 firstDecal }	Array of numObjects Objects
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
decals	32-bit	Decal { S32 dummy[5] }	Array of numDecals Decals. Note that decals are deprecated.
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
iflMaterials	32-bit	IflMaterial { S32 nameIndex, S32 materialSlot, S32 firstFrame, S32 firstFrameOffTimeIndex, S32 numFrames }	Array of numIFLs IflMaterials

GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
subShapeFirstNode	32-bit	S32	Array of numSubShapes ints representing the index of the first node in each subshape
subShapeFirstObject	32-bit	S32	Array of numSubShapes ints representing the index of the first object in each subshape
subShapeFirstDecal	32-bit	S32	Array of numSubShapes ints representing the index of the first decal in each subshape
subShapeFirstTranslucentObject	32-bit	S32	Array of numSubShapes ints representing the index of the first translucent object in each subshape
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
defaultRotations	16-bit	Quat16 { S16 x, S16 y, S16 z, S16 Z }	Array of numNodes quaternions for default node rotations
defaultTranslations	32-bit	Point3F { F32 x, F32 y, F32 z }	Array of numNodes points for default node translations
nodeRotations	16-bit	Quat16 { S16 x, S16 y, S16 z, S16 Z }	Array of numNodeRotations quaternions for node rotation keyframes (all sequences)
nodeTranslations	32-bit	Point3F { F32 x, F32 y, F32 z }	Array of numNodeTranslations points for node translation keyframes (all sequences)
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
nodeUniformScales	32-bit	F32	Array of numNodeUniformScales floats for node uniform scale keyframes (all sequences)
nodeAlignedScales	32-bit	Point3F { F32 x, F32 y, F32 z }	Array of numNodeAlignedScales points for node aligned scale keyframes (all sequences)
nodeArbScaleFactors	32-bit	Point3F { F32 x, F32 y, F32 z }	Array of numNodeArbScales points for node arbitrary scale factor keyframes (all sequences)
nodeArbScaleRots	16-bit	Quat16 { S16 x, S16 y, S16 z, S16 w }	Array of numNodeArbScales quaternions for node arbitrary scale rotation keyframes (all sequences)
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each

			buffer)
groundTranslations	32-bit	Point3F { F32 x, F32 y, F32 z }	Array of numGroundFrames points for ground transform keyframes (all sequences)
groundRotations	16-bit	Quat16 { S16 x, S16 y, S16 z, S16 w }	Array of numGroundFrames quaternions for ground transform keyframes (all sequences)
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
objectStates	32-bit	ObjectState { F32 vis, S32 frameIndex, S32 matFrame }	Array of numObjectStates ObjectStates
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
decalStates	32-bit	S32	Array of numDecalStates dummy integers for decal states
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
triggers	32-bit	Trigger { U32 state, F32 pos }	Array of numTriggers sequence triggers (all sequences)
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
details	32-bit	Detail v25- { S32 nameIndex, S32 subShapeNum, S32 objectDetailNum, F32 size, F32 averageError, F32 maxError, S32 polyCount } Detail v26+ also includes { S32 bbDimension, S32 bbDetailLevel, S32 bbEquatorSteps, S32 bbPolarSteps, F32 bbPolarAngle, bool bbIncludePoles }	Array of numDetails Details
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
meshes	ALL	Mesh	Array of numMeshes Meshes
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
names	8-bit	char[]	Array of numNames strings, stored as N characters followed by a terminating NULL for each string.
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)

alphaIn	32-bit	F32	Array of numDetails floats representing alpha-in value for each detail
alphaOut	32-bit	F32	Array of numDetails floats representing alpha-out value for each detail

Meshes

Meshes are stored in the 3 data buffers as follows:

Name	Buffer	Data Type	Description
type	32-bit	U32	Type of mesh
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)
numFrames	32-bit	S32	Number of vertex position keyframes
numMatFrames	32-bit	S32	Number of vertex UV keyframes
parentMesh	32-bit	S32	Index of this mesh's parent (usually -1 for none)
bounds	32-bit	BoxF { Point3F min, Point3F max }	Bounding box for this mesh
center	32-bit	Point3F { F32 x, F32 y, F32 z }	Bounds center for this mesh
radius	32-bit	F32	Bounding sphere radius for this mesh
numVerts	32-bit	S32	Number of vertex positions
verts	32-bit	Point3F { F32 x, F32 y, F32 z }	Array of numVerts vertex positions (all keyframes)
numTVerts	32-bit	S32	Number of UV coordinates
tverts	32-bit	Point2F { F32 u, F32 v }	Array of numTVerts UV coordinates (all keyframes)
numTVerts2	32-bit	S32	Number of 2nd UV coordinates (DTS v26+ only)
tverts2	32-bit	Point2F { F32 u, F32 v }	Array of numTVerts2 2nd UV coordinates (DTS v26+ only)
numVColors	32-bit	S32	Number of vertex color values (DTS v26+ only)
colors	32-bit	ColorI { U8 red, U8 green, U8 blue, U8 alpha }	Array of numVColors vertex colors (DTS v26+ only)
norms	32-bit	Point3F { F32 x, F32 y, F32 z }	Array of numVerts vertex normals
encodedNorms	8-bit	U8	Array of numVerts encoded normal indices
numPrimitives	32-bit	S32	Number of mesh primitives (triangles, triangle lists etc)
primitives (v24-)	16-bit	S16	Array of numPrimitives 16-bit Primitive struct data { start, numElements }
primitives (v24-)	32-bit	U32	Array of numPrimitives 32-bit Primitive struct data { maxIndex }

primitives (v25+)	32-bit	Primitive { S32 start, S32 numElements, U32 matIndex }	Array of numPrimitives Primitives
numIndices	32-bit	S32	Total number of vertex indices (all primitives)
indices (DTS v25-)	16-bit	S16	Array of numIndices vertex indices
indices (DTS v25+)	32-bit	S32	Array of numIndices vertex indices
numMergeIndices	32-bit	S32	Number of merge indices. Note that merge indices have been deprecated.
mergeIndices	16-bit	S16	Array of numMergeIndices merge indices
vertsPerFrame	32-bit	S32	Number of vertices in each keyframe (position or UV)
flags	32-bit	U32	Mesh flags
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)

If the mesh type is TSSkinMesh, the following data is stored in addition to the normal Mesh data:

Name	**Buffer**	**Data Type**	**Description**
numInitialVerts	32-bit	S32	Number of intial vert positions and normals
initialVerts	32-bit	S32	Array of numInitialVerts positions
norms	32-bit	Point3F { F32 x, F32 y, F32 z }	Array of numInitialVerts vertex normals
encodedNorms	8-bit	U8	Array of numInitialVerts encoded initial normal indices
numInitialTransforms	32-bit	S32	Number of initial transforms
initialTransforms	32-bit	MatrixF { F32 m[16] }	Array of numInitialTransforms transforms
numVertIndices	32-bit	S32	Number of vertex indices
vertIndices	32-bit	S32	Array of numVertIndices vertex indices
numBoneIndices	32-bit	S32	Number of bone indices
boneIndices	32-bit	S32	Array of numBoneIndices bone indices
numWeights	32-bit	S32	Number of weights
weights	32-bit	F32	Array of numWeights bone weights
numNodeIndices	32-bit	S32	Number of node indices
nodeIndices	32-bit	S32	Array of node indices
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)

If the mesh type is TSSortedMesh, the following data is stored in addition to the normal Mesh data:

Name	**Buffer**	**Data Type**	**Description**
numClusters	32-bit	S32	Number of clusters

clusters	32-bit	Cluster { S32 startPrimitive, S32 endPrimitive, Point3F normal, F32 k, S32 frontCluster, S32 backCluster }	Array of numClusters Clusters
numStartClusters	32-bit	S32	Number of start cluster indices
startClusters	32-bit	S32	Array of numStartClusters start cluster indices
numFirstVerts	32-bit	S32	Number of first vertex indices
firstVerts	32-bit	S32	Array of numFirstVerts first vertex indices
numNumVerts	32-bit	S32	Number of numVert counts
numVerts	32-bit	S32	Array of numVert counts
numFirstTVerts	32-bit	S32	Number of first TVert indices
firstTVerts	32-bit	S32	Array of numFIrstTVerts first TVert indices
alwaysWriteDepth	32-bit	S32	Always write depth flag
GUARD	ALL	S32, S16, S8	Buffer checkpoint (incrementing unit in each buffer)

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